MENTAL HEALTH PREDICTION SYSTEM

MINI PROJECT REPORT

OF INTERNSHIP

BACHELOR OF TECHNOLOGY
COMPUTER SCIENCE AND ENGINEERING

Submitted

To

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DECLARATION

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported and guided me throughout the development of my project, **"Mental Health Prediction System."**

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I am also grateful to my institution for providing me with the opportunity to apply my theoretical knowledge to a practical problem. This project helped me gain a better understanding of machine learning, data analysis, model evaluation, and web application development.

I would also like to acknowledge the resources, tools, and technologies that supported the development of this project. The availability of student performance data and open-source machine learning libraries helped me train and evaluate different prediction models.

I sincerely appreciate the support and encouragement of my family, friends, and everyone who motivated me throughout this journey. Their support helped me overcome challenges and complete the project while improving my technical and problem-solving skills.

Finally, I hope this project demonstrates how machine learning can contribute to the early identification of students who may benefit from wellbeing support and help students and counsellors make more informed decisions.

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ABSTRACT

Student mental wellbeing is an important part of academic life, yet students who are under strain are often identified late. Traditional approaches depend on self-reporting, periodic counselling sessions, and manual observation, which may not provide timely warning. The **Mental Health Prediction System** (web interface name: **MindPulse AI**) addresses this challenge by using machine learning to estimate a student's mental health score from everyday habits and personal indicators.

The system uses factors such as daily social media usage, number of phone unlocks, study hours, sleep hours, physical activity, self-reported stress level, age, gender, academic level, most-used platform, purpose of use, and country to predict a Mental Health Score on a scale of about 1 to 10. A dataset of 5,000 student records was cleaned, analysed, and transformed using a scikit-learn pipeline. Two machine learning approaches were trained and evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2): Linear Regression and Random Forest Regressor, the latter with and without hyperparameter tuning.

Based on the evaluation results, the default Random Forest Regressor performs best (R^2 of 0.878, MAE of 0.347 on test data) and is selected for the system. The trained pipeline is served through a FastAPI backend with strict input validation, and a web interface allows users to enter their details, obtain a predicted score with a confidence level, view a wellbeing category, receive personalised recommendations, see a counsellor alert with helpline information, and simulate how better sleep or lower screen time could change the score.

The primary objective of this project is to demonstrate how machine learning can support early awareness of student wellbeing. The system is designed as a screening and awareness tool, not as a clinical diagnostic tool, and it does not replace the judgment of qualified mental health professionals. With further improvements in dataset size, validation, and integration with campus counselling services, it could become a useful tool for student wellbeing monitoring and early support.

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CHAPTER 1: OBJECTIVES

1.1 Introduction

Mental wellbeing is an important part of a student's educational journey. However, identifying students who may be under strain is not always easy. Students often manage study pressure, irregular sleep, long hours of social media use, and limited physical activity at the same time, and the effect of these habits on their mental health may go unnoticed.

The system aims to make this process easier by using machine learning to predict a student's mental health score based on daily habits and personal indicators. Instead of relying only on occasional counselling or self-assessment questionnaires, the system analyses available information to estimate wellbeing and provide useful insights.

The main purpose of this project is to develop a simple and accessible system that helps students become aware of their wellbeing status and encourages them to seek support before problems become serious.

1.2 Objectives of the Project

The major objectives of this project are as follows:

1. Predict Student Mental Health Score

The primary objective is to estimate a student's mental health score using information such as daily social media usage, phone unlocks, study hours, sleep hours, physical activity hours, and stress level.

2. Analyze Factors Affecting Mental Health

The system examines relationships between lifestyle indicators and the mental health score. This helps identify patterns that may be associated with better or poorer wellbeing, such as the link between stress, sleep, and screen time.

3. Prepare and Process Student Data

The collected dataset is cleaned, organized, and transformed into a suitable format for machine learning. This includes checking missing values, identifying duplicate records, selecting relevant features, and encoding categorical variables.

4. Implement Multiple Machine Learning Algorithms

The project uses Linear Regression, Random Forest Regressor, to predict final mental health scores. Implementing multiple models makes it possible to compare their performance rather than depending on a single algorithm.

5. Evaluate and Select the Best Model

The models are compared using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 score. Based on the reported results, Random Forest Regressor is selected for deployment.

6. Develop a User-Friendly Web Application

The selected model is integrated into a Render application so that users can enter mental health information and obtain predictions without interacting directly with the underlying machine learning code.

7. Provide Performance Classification and Suggestions

The system classifies predicted scores into categories such as Critical Concern, At Risk, Moderate, Good, and Thriving. It also provides recommendations based on the predicted category and the user's own habits.

8. Support Early Help-Seeking

The overall objective is to give students an additional source of information that can help them recognize when they may need support. Students with low scores are encouraged to consult a counsellor, and helpline details are provided.

1.3 Expected Outcome

The expected outcome is a functional web-based application that accepts student information, predicts a mental health score, and presents the result in an understandable format along with guidance for improvement.

CHAPTER 2: TOOLS AND TECHNOLOGIES USED

2.1 Introduction

The system combines data processing, machine learning, statistical analysis, and web application development. Different tools are used at different stages, from preparing the student dataset to training prediction models and presenting their results through a web interface.

The project primarily uses Python and its machine learning ecosystem, while FastAPI and a web front-end provide the interface through which users interact with the trained model.

2.2 Python

Python is the primary programming language used for developing the project. It provides a straightforward way to handle datasets, perform calculations, train machine learning models, and integrate the prediction logic into a web application.

Its extensive collection of data science and machine learning libraries makes it suitable for building this project.

2.3 Pandas

Pandas is used for working with structured data in tabular form. It helps load, inspect, filter, and manipulate the dataset.

In this project, Pandas can be used to examine records, select relevant columns, check data quality, and organize the input features required for model training.

2.4 NumPy

NumPy supports numerical computations and operations on arrays. It is useful for handling numerical data and performing calculations required during data processing and machine learning.

2.5 Scikit-learn

Scikit-learn provides tools for data preprocessing, regression algorithms, and model evaluation.

In this project, it supports the implementation of Linear Regression and Random Forest Regressor, along with the calculation of evaluation metrics such as MAE, RMSE, and R^2 score.

It also provides utilities that can be used to prepare data consistently for training and testing.

2.6 Pydantic

Pydantic is used to define the structure of the request and response data. It validates every input, for example by rejecting negative age values, hours above 24, or unknown categories, before the data reaches the model.

2.7 Render

Render is the cloud platform configured for online deployment. The project includes a deployment file that installs the required packages and starts the application with Uvicorn.

Deployment makes the application accessible through the web, allowing users to interact with the prediction interface without having to execute the complete project locally.

2.8 Matplotlib and Seaborn

Matplotlib and Seaborn are visualization libraries that can be used to create graphs for exploratory data analysis.

The project presentation identifies histograms, box plots, scatter plots, and correlation heatmaps as part of its analysis.

These visualizations help make patterns in the dataset easier to understand and support the examination of relationships between academic indicators and final scores.

2.9 Kaggle Dataset

Kaggle is the source of the Mental Health Prediction dataset described in the project presentation.

The project uses the "Student Social Media And Mental Health Impact" dataset. It contains 5,000 student records and 13 columns covering demographics, social media usage, lifestyle habits, stress level, and the mental health score.

2.10 FastAPI and Uvicorn

FastAPI is a modern Python web framework used to build the backend of the Mental Health Prediction System. It exposes the prediction and health-check endpoints and automatically generates interactive API documentation.

Uvicorn is the server that runs the FastAPI application.

CHAPTER 3: SYSTEM FUNCTIONALITY

3.1 Introduction

The system is designed to make mental health score prediction simple and accessible. The user does not need to understand machine learning algorithms or perform calculations manually. The application handles the prediction process after the required information is entered.

The system takes selected lifestyle and personal indicators as input, processes them using a trained machine learning model, and presents the predicted score along with a wellbeing category, a confidence level, and recommendations.

3.2 Mental Health Input

The process begins when the user enters the information through the Render interface.

The selected attributes described in the project include:

- Country
- Gender.
- Age.
- Academic level (High School, Undergraduate, Graduate).
- Average daily social media usage hours.
- Daily phone unlock count.
- Daily physical activity hours.
- Average sleep hours per night.
- Stress level (Low, Medium, High, Very High).
- Main purpose of social media use.
- Most used social media platform.

3.3 Input Validation

The project includes automatic input validation. Instead of trusting the values entered by the user, the backend checks every field before prediction.

For example, age must be between 5 and 120, hours must be between 0 and 24, and daily unlocks must be between 0 and 1000. Gender, stress level, platform, and purpose must be one of the allowed values.

If any value is invalid, the system rejects the request with a clear error message (HTTP 422) instead of producing a wrong prediction.

3.4 Input Processing

Before the prediction is generated, the entered information must be converted into the format expected by the trained model.

The country is first mapped to one of the ten most common countries used in training; all other countries are grouped as "Other". The values are then arranged into a table with the same column names used during training.

This stage is important because the model expects a particular set of features. If the input structure or encoding differs from what was used during training, the prediction may be incorrect or the application may produce an error. The saved pipeline applies scaling and encoding automatically.

3.5 Mental Health Score Prediction

After the input data has been prepared, it is passed to the trained Random Forest Regressor.

The model analyzes the input features using patterns learned from the training dataset and produces an estimated mental health score. The score is limited to the range 1.0 to 10.0 and rounded to two decimal places.

The system also estimates a confidence level. Each of the 100 trees in the forest makes its own prediction, and the spread of these predictions is used to calculate confidence: the more the trees agree, the higher the confidence.

However, the predicted score is only an estimate. A student's actual wellbeing may differ because mental health is influenced by many factors that the model cannot observe.

3.6 Wellbeing Classification

Once the predicted score is generated, the application assigns it to a wellbeing category.

The categories used in the project are:

Critical Concern: Predicted score below 3.5.

At Risk (Needs Attention): Predicted score from 3.5 up to 5.0.
Moderate (Stable): Predicted score from 5.0 up to 6.5.
Good (Healthy): Predicted score from 6.5 up to 8.0.
Thriving (Excellent): Predicted score of 8.0 or above.
These thresholds are defined in the application code and are rule-based.

3.7 Personalized Suggestions

This project also provides suggestions based on the predicted performance category.

For example, if sleep is below 6.5 hours, the system suggests aiming for 7 to 8 hours of rest. If social media use is 4 hours or more per day, it suggests screen-time limits. If physical activity is below 1 hour, it suggests a daily walk or light exercise. Students in the moderate range receive stress-relief and study-break suggestions, and students in the good or thriving range receive tips to maintain their habits.

The purpose of this feature is to make the prediction more useful than a numerical score alone by connecting the result with possible next steps.

3.8 Counsellor Alert and Helplines

For students whose predicted score is below 5.0, the system strongly suggests consulting a professional counsellor. Students with a moderate score but High or Very High stress are also advised to speak with a counsellor.

A helpline window lists 24/7 support numbers for the US and Canada, the United Kingdom, India, and Australia.

3.9 What-If Habit Simulator

The application includes a what-if simulator. The user can choose to add some hours of sleep or reduce some hours of screen time, and the system recalculates the score using the same model.

The simulator shows the new predicted score and the difference from the original score. It helps users see how changes in habits are associated with changes in the predicted score. It does not guarantee the same change in real life.

CHAPTER 4: MODEL WORKFLOW

4.1 Step 1: Dataset Collection

The first stage involves collecting the "Student Social Media And Mental Health Impact" dataset.

The dataset contains 5,000 student records and 13 columns: Age, Gender, Country, Academic Level, Most Used Platform, Purpose of Use, Average Daily Usage Hours, Daily Unlocks, Study Hours, Physical Activity Hours, Sleep Hours Per Night, Stress Level, and Mental Health Score. All 12 columns other than the Mental Health Score are used as input features.

4.2 Step 2: Data Preprocessing

The collected dataset is examined and prepared before model training.

The preprocessing stage includes checking for missing values, checking duplicate records, removing unnecessary columns, and converting categorical information into numerical representations where required.

4.3 Step 3: Exploratory Data Analysis

Exploratory Data Analysis is performed to understand the distribution of student information and investigate relationships between the selected features. The project includes univariate analysis, bivariate analysis, and correlation analysis. A histogram with a density curve shows the distribution of the mental health score. A box plot compares scores across stress levels. Scatter plots examine the relationship of the score with daily usage hours and with sleep hours. A count plot shows the most-used platforms, and a correlation heatmap gives a visual summary of correlations. An outlier check using the interquartile range found 2 outliers in study hours and 22 in physical activity hours.

4.4 Step 4: Feature Selection

Skewness was calculated for the numeric features. Study hours had the highest skewness (0.44), so a log transformation was applied to it. The other numeric features were close to symmetric and were only standardized. The country column has many values, so only the ten most frequent countries were kept and all others were grouped as "Other". This reduces the number of encoded columns.

4.5 Step 5: Preprocessing Pipeline and Data Split

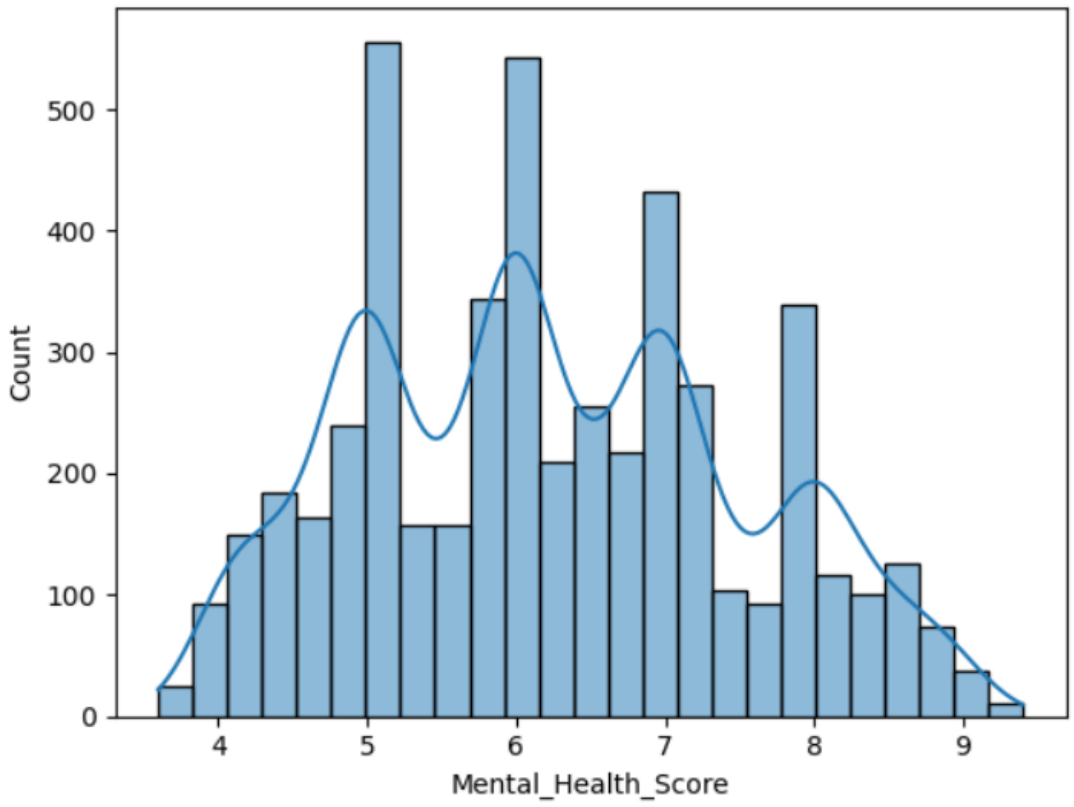
Preprocessing is built using a ColumnTransformer inside a scikit-learn Pipeline.

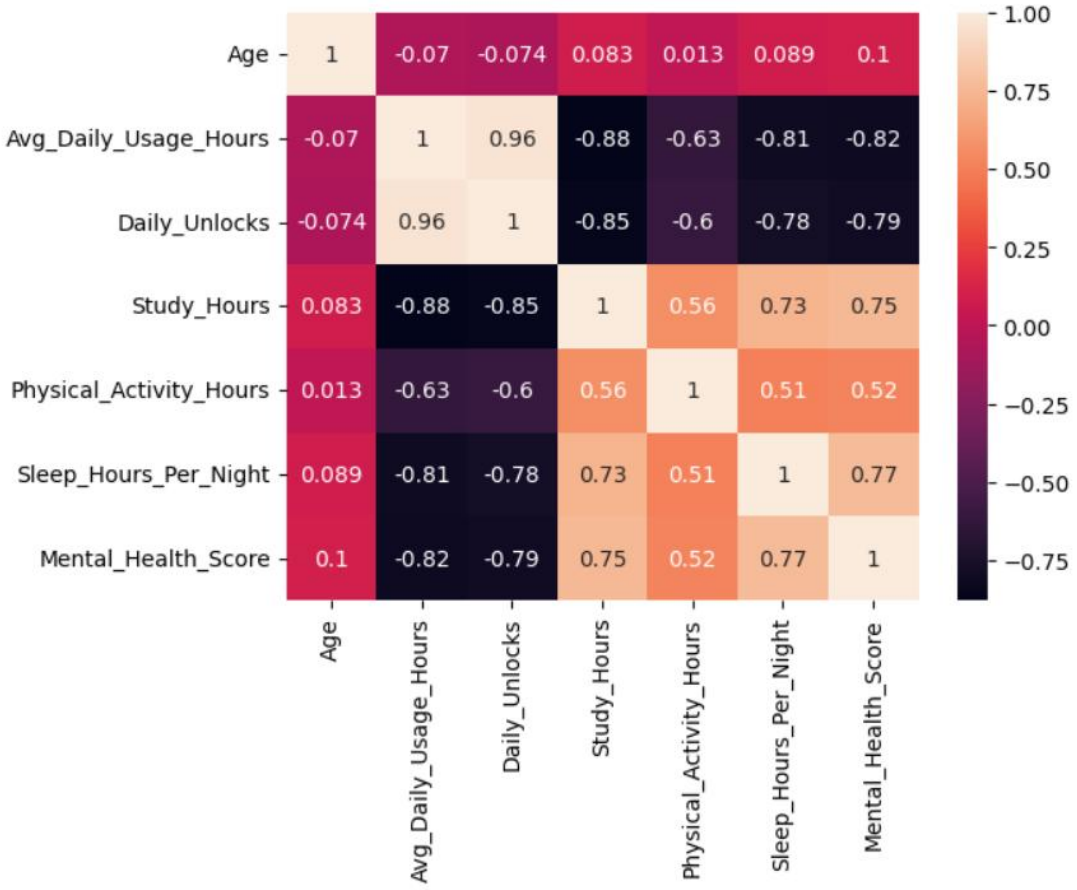
- **Study hours:** log transformation followed by standard scaling Random Forest Regressor
- **Age, daily usage hours, daily unlocks, physical activity hours, and sleep hours:** standard scaling.
- **Stress level:** ordinal encoding in the order Low, Medium, High, Very High
- Gender, academic level, platform, purpose of use, and grouped country: one-hot encoding.

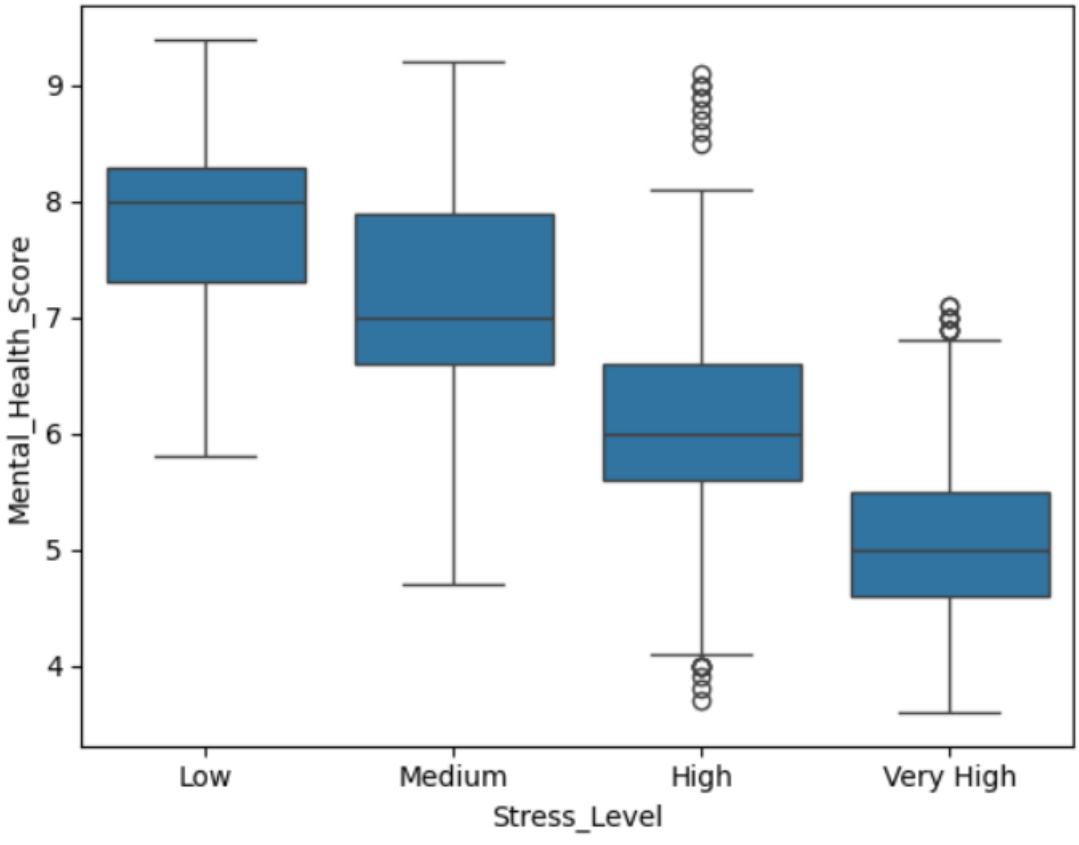
4.6 Step 7: Model Training

Three regression models are used to learn the relationship between student attributes and the mental health score.

- Linear Regression (baseline)
- Random Forest Regressor (default settings, 100 trees)
- Random Forest Regressor (tuned using RandomizedSearchCV with 15 combinations and 5-fold cross-validation)




Age
Avg_Daily_Usage_Hours
Daily_Unlocks
Study_Hours
Physical_Activity_Hours
Sleep_Hours_Per_Night
Mental_Health_Score
1 -0.07 -0.074 0.083 0.013 0.089 0.1
-0.07 1 0.96 -0.88 -0.63 -0.81 -0.82
-0.074 0.96 1 -0.85 -0.6 -0.78 -0.79
0.083 -0.88 -0.85 1 0.56 0.73 0.75
0.013 -0.63 -0.6 0.56 1 0.51 0.52
0.089 -0.81 -0.78 0.73 0.51 1 0.77
0.1 -0.82 -0.79 0.75 0.52 0.77 1
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75


Mental_Health_Score
9
8
7
6
5
4
Low
Medium
High
Very High
Stress_Level

4.7 Step 7: Model Evaluation

After training, the models are evaluated on the test data using three metrics.

- **Mean Absolute Error (MAE):** Measures the average absolute difference between predicted and actual scores. Lower values indicate smaller average errors.
- **Root Mean Squared Error (RMSE):** Measures prediction error while giving greater influence to larger errors. Lower values are preferred.
- **R^2 Score:** Measures how well the model explains variation in the target variable relative to a baseline. Higher values generally indicate better performance.

The results obtained are:

Model	MAE	RMSE	R^2 Score
Linear Regression	0.536	0.676	0.740
Random Forest (Default)	0.347	0.464	0.878
Random Forest (Tuned)	0.369	0.487	0.865

The default Random Forest performs best on all three metrics and is selected for deployment.

The default Random Forest Regressor is selected because it achieves the lowest MAE and RMSE and the highest R^2 score among the evaluated models. Its MAE of 0.347 indicates an average absolute prediction error of approximately 0.35 points on a score range of about 3.6 to 9.4. Its R^2 score of 0.878 also indicates that the model explains most of the variation in mental health scores, although some variation remains unexplained.

4.8 Step 8: Deployment

After model selection, the trained Random Forest Regressor is integrated into the Render web application.

CHAPTER 5: KEY FEATURES

5.1 Machine Learning-Based Score Prediction

The primary feature of student performance prediction system is its ability to estimate a final mental health score using selected student attributes.

5.2 Automatic Input Validation

The system checks every input for valid ranges and allowed values, and rejects invalid data with a clear error message.

5.3 Real-Time Prediction

After the required information is submitted, the application generates the predicted score and displays the result through the interface within moments, because the model is loaded once at start-up.

5.4 Confidence Estimation

The system reports a confidence percentage based on how closely the 100 decision trees agree with each other.

5.5 Wellbeing Classification

The predicted score is assigned to one of five wellbeing categories: Critical Concern, At Risk, Moderate, Good, and Thriving.

5.6 Personalized Recommendations

The system provides recommendations based on the predicted category and the user's own sleep, screen time, and activity..

5.7 Counsellor Alert and Helplines

Students with low predicted scores are encouraged to consult a counsellor, and 24/7 helpline numbers are provided for four regions.

5.8 What-If Habit Simulator

The simulator lets users test how more sleep or less screen time could change their predicted score.

CHAPTER 6: PERFORMANCE AND EVALUATION

6.1 Evaluation Metrics

Mean Absolute Error (MAE)

Mean Absolute Error measures the average absolute difference between actual scores and predicted scores.

It provides a straightforward interpretation of how far predictions are from the actual values on average.

A lower MAE indicates that the model produces smaller average absolute errors.

For the Mental Health Prediction System, the default Random Forest Regressor achieves an MAE of 0.347, the lowest among the three evaluated models.

Root Mean Squared Error (RMSE)

Root Mean Squared Error is calculated by squaring the prediction errors, averaging them, and taking the square root of the result.

Because the errors are squared, larger errors have a greater influence on the final metric.

A lower RMSE generally indicates better predictive performance, particularly when large prediction errors are a concern.

The default Random Forest Regressor achieves an RMSE of 0.464, compared with 0.676 for Linear Regression and 0.487 for the tuned Random Forest.

R^2 Score

The R^2 score measures how well a model explains the variation in the target variable relative to a baseline prediction.

Higher values generally indicate that the model explains more of the observed variation. An R^2 score should not be interpreted as a percentage accuracy.

The default Random Forest Regressor achieves an R^2 score of 0.878 on test data, which is higher than the scores of the other two models.

6.2 Model Performance Comparison

The results reported in the project presentation are summarized below.

Model	Training R	MAE	RMSE	R^2 Score
Linear Regression	0.724	0.536	0.676	0.740
Random Forest (Default)	0.981	0.347	0.464	0.878
Random Forest (Tuned)	0.955	0.369	0.487	0.865

Linear Regression

Linear Regression records an MAE of 0.536 and an RMSE of 0.676. Its R^2 score is 0.740.

It provides a baseline for the experiment, but its results are weaker than those of both Random Forest models. Its training and test scores are close (0.724 and 0.740).

Random Forest Regressor (default)

Random Forest records an MAE of 2.99, an RMSE of 3.72, and an R^2 score of 0.326.

It achieves the lowest error values and the highest R^2 score among the evaluated models.

Random Forest Regressor (tuned)

The tuned Random Forest records an MAE of 0.369, an RMSE of 0.487, and a test R^2 score of 0.865.

Tuning reduced the gap between training and test scores (0.955 against 0.865), but test performance was slightly weaker than the default Random Forest.

6.3 Selection of the Final Model

Random Forest is selected because it performs best across all three reported evaluation metrics.

Its MAE is lower than that of Linear Regression and the tuned Random Forest, meaning it has a smaller average absolute prediction error on the test data.

Its RMSE is also the lowest, indicating better performance under a metric that gives greater weight to large errors.

Finally, its R^2 score is the highest, suggesting that it explains more of the variation in mental health scores than the other models in this comparison.

These results justify selecting the default Random Forest for the current implementation.

6.4 Interpretation of the Results

The evaluation results demonstrate that the three algorithms do not perform equally on the selected dataset.

Random Forest provides the strongest results among the models tested. Its R^2 score of 0.878 shows that the model explains a large share of the variation in scores. This also suggests that the relationship between habits and mental health score is non-linear.

The result should be interpreted carefully. It does not mean the system predicts scores with 87.8% accuracy. Instead, it indicates that the model explains approximately 87.8% of the variation in target scores on the test data.

The MAE of 0.347 indicates an average absolute prediction error of approximately 0.35 points on a score scale of roughly 1 to 10.

The default Random Forest scores 0.981 on training data and 0.878 on test data. This gap shows some overfitting, which is common for unpruned forests.

These metrics are meaningful only in relation to the data and evaluation procedure used. Additional validation is required to determine how reliably the model performs on new students.

6.5 Overall Evaluation

The project demonstrates a complete machine learning workflow involving data preparation, algorithm comparison, model evaluation, and application deployment. The reported metrics support selecting Random Forest over the other tested algorithms. At the same time, the results indicate that there is room for improvement.

CHAPTER 7: LIMITATIONS

7.1 Limited Dataset Size

The project uses a dataset containing 4,998 usable student records after cleaning.
While the dataset is sufficient to demonstrate the machine learning workflow, it may not capture the full diversity of students, institutions, cultures, and living conditions.

7.2 Not a Clinical Tool

The predicted score is a statistical estimate. It is not a medical or psychological diagnosis and must not replace advice from a qualified professional.

7.3 Overfitting

The selected model reaches an R^2 score of 0.981 on training data but 0.878 on test data. This indicates that it fits the training data more closely than unseen data.

7.4 Dependence on Input Data Quality

The prediction depends on the information entered into the application. Stress level, sleep hours, and social media usage are self-reported. If these values are incorrect or inconsistent, the resulting prediction may be affected. The mental health score in the dataset may also be subjective.

7.5 Limited Indicators

The model uses a selected set of student attributes. However, mental health can be influenced by many other factors, including family situation, relationships, health, financial pressure, and personal events, which are not part of the data.

7.6 Correlation, Not Causation

The model learns associations in the data. The what-if simulator shows how the model's output changes, not a guaranteed effect on a real person's wellbeing.

7.7 Rule-Based Recommendations

The wellbeing thresholds, recommendations, and helpline list are written manually in the code. They have not been validated by mental health professionals and need manual updates.

CHAPTER 8: CONCLUSION AND FUTURE WORK

8.1 Conclusion

This project successfully demonstrates the application of Machine Learning in predicting student mental health scores from social media usage and lifestyle habits.

The project involved data collection, preprocessing, exploratory data analysis, feature engineering, model training, model evaluation, and web deployment.

Among the implemented algorithms, the default Random Forest Regressor achieved the best performance (R^2 of 0.878 and MAE of 0.347) and was selected for deployment.

The developed FastAPI application with the MindPulse AI interface provides a simple and interactive way through which users can enter their details and obtain a predicted score, category, confidence, recommendations, and counsellor guidance instantly.

8.2 Future Scope

The project can be further enhanced in several ways:

- Use larger, verified, real-world datasets covering more ages and countries.
- Try advanced models such as Gradient Boosting, XGBoost, or LightGBM.
- Calibrate the confidence level using statistical methods.
- Integrate with college counselling services and student support systems.
- Add user accounts so students can track their progress over time.
- Develop Android and iOS applications.
- Add automated tests, Docker packaging, and monitoring.
- Validate recommendations with mental health professionals.

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APPENDIX

Appendix A: Dataset

Student Social Media And Mental Health Impact dataset containing 5,000 records and 13 columns. 4,998 records were used. Twelve input features and one target were used.

Appendix B: Data Preprocessing

The dataset was checked for missing values and duplicate records. Relevant features were selected, and categorical variables were converted into numerical values where required.

Appendix C: Machine Learning Models

Regression, Random Forest Regressor (default), Random Forest Regressor (tuned with RandomizedSearchCV)

Appendix D: Model Evaluation

Model	MAE	RMSE	R^2 Score
Linear Regression	0.536	0.676	0.740
Random Forest	0.347	0.464	0.878
Random Forest (tuned)	0.369	0.487	0.865

Appendix E: Confidence Calculation

Confidence (%) = $100 \times (1 - \sigma / 2.5)$, limited to between 50% and 99%, where σ is the standard deviation of the predictions of the 100 individual trees.

Appendix F: Tools and Technologies

The project uses Python, Pandas, NumPy, Scikit-learn, Pydantic, Matplotlib, Seaborn, and Render. Kaggle was used as the dataset source.

Appendix G: Future Improvements

Future enhancements may include larger datasets, advanced models, explainability of predictions, calibrated confidence, integration with campus counselling services.